

A Feature Extraction Approach Over Workpiece Point Clouds for Robotic Welding

Yuankai Zhang^{ID}, Yusen Geng^{ID}, Xincheng Tian^{ID}, Fuquan Zheng^{ID}, Yong Jiang^{ID}, and Min Lai^{ID}

Abstract—The programming of welding trajectories hinders the development of welding robots. With the development of machine vision, the welding path planning based on 3D vision can eliminate the teaching and programming work of the robot. However, the current weld extraction research relies on the geometric information of the workpiece, resulting in the low generality and robustness of the existing methods. The edge and corner features of the point cloud are the basis for extracting welds. Therefore, in view of the above problems, this paper proposes a method for edge and corner extraction of workpieces applied to 3D vision welding. First, the LOBB feature descriptor is proposed to build a feature space for the workpiece point cloud. Then, to enhance the robustness of feature clustering, nonlinear activation is performed on the feature description. Finally, a hierarchical clustering method based on K-means is designed to achieve the extraction of crease points, boundary points, and corner points. Experiments show that the method in this paper can not only extract the edges of various workpieces completely and with low noise, but also can identify corners efficiently, which is overall better than the existing methods.

Note to Practitioners—Welding trajectory programming has been a bottleneck in the development of welding robots. However, advancements in machine vision have opened up new possibilities for welding path planning using 3D vision, which can eliminate the need for manual teaching and programming of robots. Unfortunately, existing methods for weld extraction from point clouds rely heavily on geometric information, resulting in limited applicability and robustness. To address these challenges, this paper proposes a novel method for extracting edge and corner features from workpieces using 3D vision in welding applications. The key contribution is the introduction of the LOBB feature descriptor, which creates a feature space for the workpiece's point cloud. By applying nonlinear activation to the feature description, the robustness of feature clustering is enhanced. Additionally, a hierarchical clustering method based on K-means is designed

to extract crease points, boundary points, and corner points. Experimental results demonstrate that the proposed method can effectively extract edges from various workpieces with minimal noise and efficiently identify corners. Overall, the method outperforms existing approaches in terms of completeness and accuracy of edge extraction. This advancement holds significant potential for practitioners in the field of welding robotics, as it reduces the programming complexity and improves the reliability of welding robots in real-world applications.

Index Terms—Weld seam extraction, point cloud feature extraction, LOBB feature descriptor, K-means.

I. INTRODUCTION

TRADITIONAL manual welding has disadvantages such as harsh working environments, high skill requirements of workers, and low efficiency. Therefore, to overcome the shortcomings of manual welding, welding robots began to be widely used in machining, construction engineering, aerospace, automobile manufacturing, pipeline transportation, and microelectronics [1]. However, the complex and tedious programming [2], [3] and teaching work [4], [5] affects the efficiency of welding robots. With the development of machine vision, welding robots based on vision guidance started to appear in order to improve the efficiency of welding robots further.

The 2D vision sensor applied to weld seam tracking mainly consists of an industrial camera and a line laser projector [6]. Due to the limited detection area of the line laser, the welding path planning method based on the line laser vision sensor needs to perform rough teaching according to the welding seam of the workpiece and then process the two-dimensional image during the robot welding process to realize real-time welding seam tracking [7], [8], [9]. Therefore, arc interference in the welding process significantly impacts this type of method [10], [11], [12]. Moreover, this kind of method still cannot avoid teaching the welding trajectory and only limitedly improves the welding efficiency.

With the development of 3D vision, the point cloud information of the workpiece surface can be obtained by scanning workpieces with 3D vision sensors [13], [14], [15]. Point clouds can accurately describe the geometric features of the surface of an object. Therefore, more and more researchers perform weld feature extraction based on point clouds [16], [17], [18]. Yang et al. designed a 3D vision sensor based on a digital light processing (DLP) projector to collect the point cloud of the workpiece and used the geometric features of the V-shaped weld to achieve weld extraction [19]. Patil et al. first clustered the workpiece surface and then extracted spatial I-type butt welds according to the offset direction of the

Manuscript received 18 October 2023; accepted 19 December 2023. Date of publication 1 January 2024; date of current version 13 January 2025. This article was recommended for publication by Associate Editor C. Clévy and Editor J. Yi upon evaluation of the reviewers' comments. This work was supported in part by the National Natural Science Foundation of China under Grant U20A20201, in part by the Shandong Provincial Key Research and Development Program under Grant 2022CXGC010101 and Grant 2022CXGC020204, and in part by the Taishan Industry Leading Talent Project. (Corresponding authors: Xincheng Tian; Fuquan Zheng.)

Yuankai Zhang, Yusen Geng, Xincheng Tian, Yong Jiang, and Min Lai are with the Center for Robotics, School of Control Science and Engineering, Shandong University, Jinan, Shandong 250061, China, and also with the Engineering Research Center of Intelligent Unmanned System, Ministry of Education, Jinan, Shandong 250061, China (e-mail: zyk_sdu@163.com; gys_sdu@163.com; txch@sdu.edu.cn; 202134859@mail.sdu.edu.cn; 202114769@mail.sdu.edu.cn).

Fuquan Zheng is with the Institute of Automation, Shandong Academy of Sciences, Jinan 250014, China (e-mail: zhengfq0429@126.com).

This article has supplementary downloadable material available at <https://doi.org/10.1109/TASE.2023.3345868>, provided by the authors.

Digital Object Identifier 10.1109/TASE.2023.3345868

1558-3783 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.

See <https://www.ieee.org/publications/rights/index.html> for more information.

boundary point neighborhood center [20]. By improving the RANSAC algorithm, Geng et al. realized the efficient fitting of the multi-plane of the point cloud of the medium and thick plate workpiece, then used the intersection lines of the surfaces to extract the weld seam, and finally realized the welding path planning on the three types of workpieces [21]. Therefore, the welding path planning based on 3D vision can overcome the shortcomings of 2D vision, avoid the teaching work of the robot, and further improve the efficiency of the welding robot.

However, through investigation, we found that the current workpiece point cloud welding seam extraction methods have the following problems:

- 1) It mainly depends on the overall geometric characteristics of the specific workpiece, so the generality is low;
- 2) There are multiple threshold adjustments for weld extraction rules, resulting in low robustness.

The main task of visual inspection in industrial scenarios is to extract application-related features. And, edge and corner features are widely used to describe the geometric information of objects [22]. In particular, in the welding scene, the weld seam is generally located at the edge of the workpiece, and the endpoint of the weld seam is generally the corner of the workpiece. Therefore, in order to overcome the shortcomings of traditional weld seam extraction methods based on specific workpiece geometric information, this paper proposes a generic idea based on point cloud feature extraction of welding seam feature extraction: First, extract the edge and corner features of the workpiece and then screen out the welds according to the welding requirements.

In order to realize the extraction of edge and corner features of welding workpieces, the main innovations of this paper are as follows:

- 1) A robust point cloud feature descriptor is designed to describe creases, boundaries, and corners;
- 2) To optimize the feature distribution, a nonlinear activation feature description method is proposed;
- 3) A hierarchical clustering algorithm based on K-means is proposed, which robustly realizes the edge and corner extraction of the workpiece;
- 4) This method is suitable for workpieces with various surface curvatures and is verified experimentally on three kinds of workpieces.

The organization of this paper is as follows: Section II summarizes the current research on point cloud-based 3D feature extraction. Section III proposes a LOBB-based point cloud feature extraction method. Section IV introduces the experimental system and then designs comparative experiments for evaluating our method. Section V summarizes the work of this paper.

II. RELATED WORK

3D feature extraction based on point clouds has consistently been a research focal point. Unlike the pixel matrix found in traditional two-dimensional images, point cloud data is unordered and discrete. Consequently, performing convolution operations directly on point cloud data is challenging [23]. Simultaneously, points within a point cloud can only be effectively characterized when they constitute a meaningful subset alongside neighboring points. Therefore, the foundation

of point cloud feature extraction lies in computing the feature descriptor within the relevant neighborhood space of a point and subsequently mapping it to the point cloud feature space. Depending on the varying descriptions of point cloud features, feature extraction methods can be categorized into two types: those based on single-point features and those based on neighborhood features.

The methods based on single-point features use single-point feature descriptions such as the curvature and normal vector of the point cloud for feature extraction [24], [25]. Chen et al. initially extracted candidate feature points based on changes in the normal vectors within the local neighborhood. They then employed an anisotropic L1-median normal filter to refine these feature points, facilitating the extraction of feature lines from noisy point clouds [26]. Zhou et al. introduced a dynamic Top-K selection strategy aimed at simplifying the fitting plane for key points. This approach improved the accuracy of normal calculations for sharp corner points, thereby enabling the extraction of edges from complex shapes [27]. The single-point feature does not have a description of the neighborhood point distribution when the feature space is constructed, so the classification based on the single-point feature of the point cloud is easily affected by the density and roughness of the point cloud.

The methods based on neighborhood features use the point cloud's neighborhood point distribution state for feature extraction [28], [29]. Yang et al. calculated the L1-median as the nearest neighbor center point. They then introduced K+1 neighbors to calculate the feature description and finally realized the extraction of subtle feature points based on Poisson distribution [30]. Tian et al. used the α -shape operator to robustly extract the line segments out of the building after projecting the point cloud to the plane [31]. Ahmed et al. evaluated the symmetry of the point cloud neighborhood and achieved edge detection based on the point cloud density [32]. He et al. proposed an edge point extraction method based on concurrent Delaunay triangular meshes (CDTMs), which adds the vertex angles of CDTMs of neighborhood points of common data points together as the evaluation index [33]. The problem with this type of method at present is that the mechanism of feature description cannot be described from geometric principles, which leads to complex feature clustering.

Based on the experience of the researchers, this study aims to propose a feature descriptor with a precise mechanism and construct a feature space with a clear distribution. On this basis, the machine learning method is used for clustering. Finally, the edge and corner features of various welding workpiece point clouds can be robustly extracted.

III. ALGORITHM

This paper proposes a local oriented bounding box descriptor whose feature descriptions of face points, crease points, and corner points are combined with the geometric features of the minimum bounding box of a fixed radius neighborhood. In order to make the feature points more prominent, we activate the feature nonlinearly. Finally, the unsupervised K-means method is used for multi-layer clustering in the feature space, which realizes the extraction of the corners and edges of the workpiece.

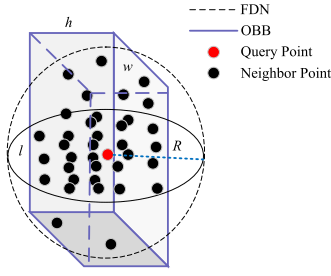


Fig. 1. Local Oriented Bounding Box (LOBB).

A. Local Oriented Bounding Box

OBB (Oriented Bounding Box) in three-dimensional space is the smallest circumscribed cuboid that encloses the discrete point set [34]. The main axis direction of OBB is aligned with the direction of the scattered point distribution, which can accurately describe the spatial distribution state of a set of discrete points. As shown in Fig. 1, the FDN (Fixed Distance Neighbor) is solved for any point in the workpiece point cloud with R as the radius, and a set of neighbor points $P = \{p_1, p_2, p_3, \dots, p_k | p_i \in R^3\}$ is obtained. Create an OBB with length l , width w , and height h for set P , where $l \geq w \geq h$. As can be seen from the figure, the OBB closely surrounds discrete points, and it is a regular cuboid, which can be described mathematically. Therefore, this study believes that the geometric features of the query point's neighborhood OBB can describe the point's neighbor features, which will be further used as the basis for feature extraction. In this paper, the OBB established on the FDN of a certain point is called the LOBB (Local Oriented Bounding Box), and the solution steps are as follows:

- 1) Denote the three-dimensional data components of the set P as three vectors $\vec{X} = (x_1, x_2, \dots, x_k)$, $\vec{Y} = (y_1, y_2, \dots, y_k)$, $\vec{Z} = (z_1, z_2, \dots, z_k)$, and solve the mean for each dimension:

$$\bar{x} = \frac{\sum_{i=1}^k x_i}{k}, \bar{y} = \frac{\sum_{i=1}^k y_i}{k}, \bar{z} = \frac{\sum_{i=1}^k z_i}{k} \quad (1)$$

- 2) Solve the covariance between every two dimensions, taking the covariance formula between $\vec{X}$ and $\vec{Y}$ as an example:

$$\text{cov}(\vec{X}, \vec{Y}) = \frac{\sum_{i=1}^k (x_i - \bar{x})(y_i - \bar{y})}{k - 1} \quad (2)$$

- 3) The covariance matrix C between the three-dimensional data is solved, and the solution formula is shown in (3). Solve the covariance matrix C to get the eigenvalue $\{\lambda_1, \lambda_2, \lambda_3\}$ and the eigenvector $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ corresponding to the eigenvalue, where $\lambda_1 \geq \lambda_2 \geq \lambda_3$.

$$C = \begin{bmatrix} \text{cov}(\vec{X}, \vec{X}) & \text{cov}(\vec{X}, \vec{Y}) & \text{cov}(\vec{X}, \vec{Z}) \\ \text{cov}(\vec{Y}, \vec{X}) & \text{cov}(\vec{Y}, \vec{Y}) & \text{cov}(\vec{Y}, \vec{Z}) \\ \text{cov}(\vec{Z}, \vec{X}) & \text{cov}(\vec{Z}, \vec{Y}) & \text{cov}(\vec{Z}, \vec{Z}) \end{bmatrix} \quad (3)$$

- 4) As illustrated in Fig. 2, we employ the eigenvector $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ as the basis for establishing a new coordinate system, thereby rotating the original coordinate system of the set P into the $X_1Y_1Z_1$ coordinate system. After this transformation, a bounding box is constructed along the X_1 -axis, Y_1 -axis, and Z_1 -axis. At this point, the

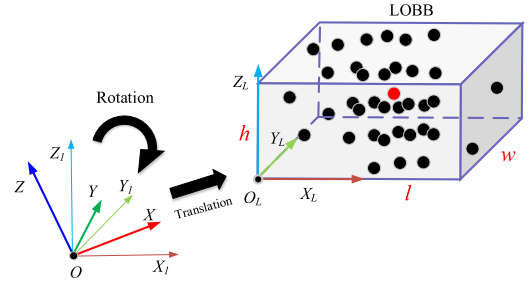


Fig. 2. LOBB construction principle.

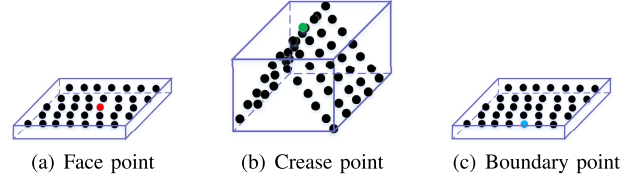


Fig. 3. LOBBs of different types of points.

dimensions of the set P along the three coordinate axes correspond to the length (l), width (w), and height (h) of the LOBB.

- 5) Lastly, in an effort to streamline subsequent calculations, we opt to use the corner point $O_L(x_{1min}, y_{1min}, z_{1min})$ from the distribution of P in the $X_1Y_1Z_1$ coordinate system as the new coordinate origin. This allows us to transform $X_1Y_1Z_1$ into the local coordinate system $X_LY_LZ_L$. The $X_LY_LZ_L$ coordinate system is labeled in Fig. 2.

B. LOBB Feature Descriptor

1) *Edge Feature Description*: The edge refers to the part where the properties of the surface of the object are abruptly changed. This paper divides edge points into crease points and boundary points according to different attribute changes. Crease points are points that reflect changes in the shape of an object. Boundary points are the furthest points of the point cloud that are no longer continuous in a certain direction. In the robot welding system based on 3D vision, the 3D camera mainly focuses on the welding seam area, and some concave and convex features of the edge of the workpiece will be lost. Therefore, in order to extract the workpiece features more completely, the edge extraction in this study includes crease points and boundary points.

Fig. 3 is a schematic diagram of different types of points in the point cloud to establish LOBB, in which the red point is a face point, the green point is a crease point, and the blue point is a boundary point. So, Fig. 3(a), Fig. 3(b), and Fig. 3(c) are the LOBB schematic diagrams of face points, crease points, and boundary points, respectively.

Comparing Fig. 3(b) with Fig. 3(a) and Fig. 3(c), respectively, it can be seen that the concave-convex changes in the neighborhood of the crease point are significant. So, the LOBB of the face and boundary points is flatter than the LOBB of the crease point. This feature can separate out the crease points. Comparing Fig. 3(c) with Fig. 3(a) and Fig. 3(b), respectively, we can see that the boundary point is located at the edge of the neighbor. Looking down from above, the boundary points are closer to the boundary of the upper surface of the bounding

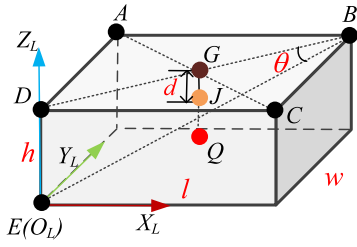


Fig. 4. LOBB feature descriptor solution schematic diagram.

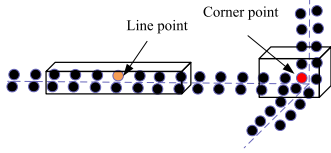


Fig. 5. LOBB feature description of corners.

box than the face and crease points. This feature description can highlight boundary points. To quantify the above features, we propose the flatness f and the center offset d .

As shown in Fig. 4, starting from the vertex B of the upper surface, we draw the diagonal of the upper surface and the cuboid diagonal, forming a right triangle BDE with the ED side. It can be found that the size of the angle θ can describe the flatness of the bounding box, so the formula for the flatness f is as follows:

$$f = \tan\theta = \frac{h}{\sqrt{l^2 + w^2}} \quad (4)$$

The solution of the center offset is based on the $X_L Y_L Z_L$ coordinate system, as shown in Fig. 4. The query point Q is any point in the point cloud, and its coordinate is set as (x_Q, y_Q, z_Q) . Point J is the projection of point Q on the upper surface of the bounding box, and point G is the center point of the upper surface. The coordinates of the bounding box vertices A, B, C, D , and E are known in the local coordinate system.

The coordinates of point G can be solved by (5), and its coordinate is $(\frac{l}{2}, \frac{w}{2}, h)$.

$$G = \frac{A + B + C + D}{4} \quad (5)$$

In the cuboid, DE and QJ are parallel, so there is the relationship of (6).

$$\frac{\overrightarrow{ED}}{\|\overrightarrow{ED}\|} = \frac{\overrightarrow{QJ}}{\|\overrightarrow{QJ}\|} \quad (6)$$

According to (6), we can solve the coordinate of point J as (x_Q, y_Q, h) . Then the formula for the center offset d can be obtained as follows:

$$d = \|\overrightarrow{GJ}\| = \sqrt{(x_Q - \frac{l}{2})^2 + (y_Q - \frac{w}{2})^2} \quad (7)$$

2) *Corner Feature Description*: Corner points are the points where edge lines meet. On the extracted edge line point cloud, LOBB is established for all points respectively. Fig. 5 shows that yellow represents a line point, and red represents a corner point. We found that the line points are distributed in bands,

so the length of the LOBB is much larger than the height and width, but the length, width, and height of the LOBB at the corner points are relatively uniform. We use the standard deviation s of the side lengths of the bounding box to describe the above features. For ease of understanding, in this paper, we call s the corner point entropy. Its formula is shown in (8).

$$m = \frac{l + w + h}{3}, s = \sqrt{\frac{(l - m)^2 + (w - m)^2 + (h - m)^2}{3}} \quad (8)$$

C. Feature Extraction Based on K-Means

The workpiece point cloud is divided into non-boundary points, boundary points according to the center offset, and face points and crease points according to the flatness. The edge point cloud can be divided into corner points and line points based on the corner entropy. The above analysis shows that all feature clustering problems in this paper are binary classification problems of one-dimensional data. And K-means is an unsupervised learning clustering algorithm that performs clustering by iterating the cluster centers until the distance of the objects within the class to the center is minimized [35]. For such problems, K-means has high clustering efficiency and is suitable for industrial scenarios. Therefore, this paper is based on the K-means method for clustering in the feature space. First, in order to optimize the effect of clustering, we perform nonlinear activation on the features and then propose a hierarchical clustering algorithm based on K-means to achieve point cloud segmentation.

1) *Non-Linear Activation*: Due to the non-uniformity of point cloud quality and workpiece surface curvature, many intermediate feature values will be located at the classification boundary during the feature calculation process. This will affect the effect of K-means clustering. In order to optimize the clustering effect, we introduce a nonlinear function to activate the feature vector. Activated features tend to be more polarized, which reduces the influence of intermediate feature values. After many experiments, we find that the input features are well polarized due to the bilateral nature of the $\tanh$ function. Therefore, this paper finally chooses the $\tanh$ function as the activation function, and the function formula is (9).

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}, (-2 \leq x \leq 2) \quad (9)$$

The activation process of the feature is as follows: We set one of the normalized feature vectors as $T = [t_1, t_2, t_3, \dots, t_n]$. First, T_1 is obtained by linearly projecting T into $[-2, 2]$, then T_1 is substituted into the above formula for activation to get T_2 , and finally, the activated T_2 is normalized to obtain T' .

This paper uses the flatness feature's nonlinear activation as an example to illustrate the effect. As shown in Fig. 6, this is a point cloud of the intersection of two tubes captured by a 3D camera. According to the weld width, we calculate the LOBB feature descriptor with a radius of 2.5mm for the workpiece point cloud and normalize the obtained feature vector. As shown in Fig. 7, the upper curve is the original feature distribution, where Cluster1 and Cluster2 are the clustering results of K-means, and the black points on the vertical axis are the cluster centers. The extraction result of

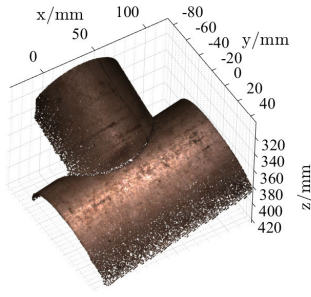


Fig. 6. The point cloud of the intersecting circular tube workpiece.

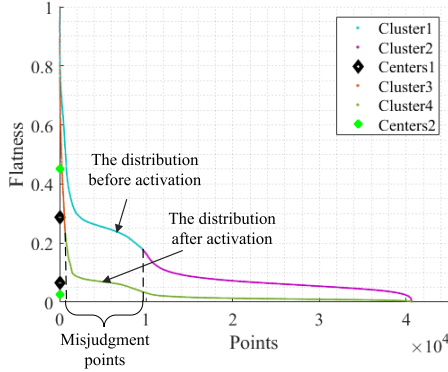


Fig. 7. Cluster analysis before and after feature activation.

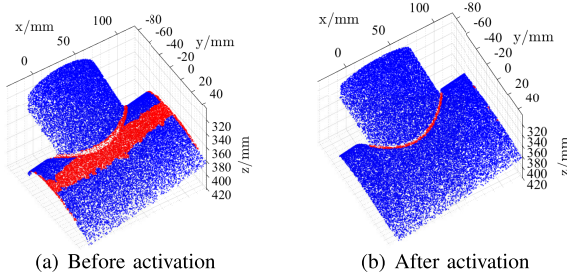


Fig. 8. Feature extraction results comparison before and after activation.

the crease points at this time is shown in Fig. 8(a), where red represents the crease points. It can be seen that many face points are misjudged as crease points, which indicates that a large number of intermediate feature values are divided into Cluster1. The curve below is the feature distribution after nonlinear activation, of which Cluster3 and Cluster4 are the clustering results of K-means, and the green points on the vertical axis are the cluster centers.

The feature distribution after activation is more polarized. Viewed on the y -axis, the distances between the resulting cluster centers become larger, which indicates that the influence of intermediate feature values is reduced. From the analysis on the x -axis, the dividing line of the second clustering is further to the left, and the part intercepted by the two dividing lines corresponds to the misjudgment points. Fig. 8(b) is the extraction result at this time. It can be seen that the segmentation of the crease points is accurate. Therefore, the comparison shows that nonlinear activation can optimize clustering results.

2) *Hierarchical K-Means Clustering*: For the workpiece point cloud, the method in this paper needs to perform

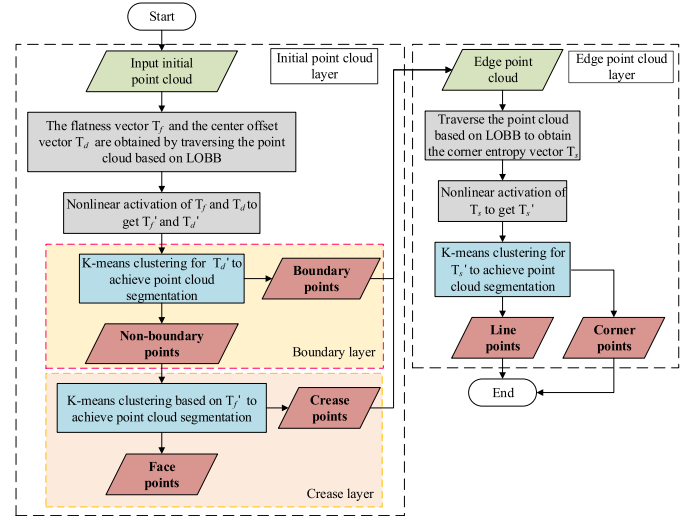


Fig. 9. Flow chart of hierarchical clustering algorithm based on K-means.

K-means clustering for each feature vector in the feature space and finally realize the segmentation of face points, crease points, boundary points, and corner points. This paper proposes a hierarchical clustering algorithm based on K-means by organically combining the three clustering processes.

The specific process of the algorithm is shown in Fig. 9. The algorithm is divided into an initial point cloud layer and an edge point cloud layer according to the input. In the initial point cloud layer, first, traverse the initial point cloud to calculate the center offset and flatness and form the flatness vector T_f and the flatness vector T_d . The nonlinear activation of the vectors gets T'_f and T'_d . Then, based on the K-means clustering output of these two features, we divide it into the boundary layer and crease layer. Among them, the boundary layer is clustered based on the center offset vector T'_d , and the crease layer is clustered based on the flatness vector T'_f . In this algorithm, the boundary layer is performed first, and then the crease layer is performed. We get boundary points and crease points merged as an edge point cloud and input to the next layer for processing. In the edge point cloud layer, the edge point cloud is first traversed, the corner point entropy is calculated for each point based on LOBB, and the feature vector T_s is formed and activated to obtain T'_s . Then perform K-means clustering on T'_s to get the corner points. Each layer of the clustering algorithm is binary classification, so the K value of each layer of K-means clustering in the algorithm is 2.

IV. EXPERIMENTS

A. Experimental System

The robot welding system relied on upon this study is shown in Fig. 10. The system is mainly composed of a 3D camera, a robot, a computer, and welding tools. Its purpose is to obtain the characteristics of the welding seam by processing point cloud data and then automatically plan the welding path to achieve programming-free and teach-free robotic welding. The camera fixed at the robot's end is a DLP area array structured light camera, which has the characteristics of high precision, fast acquisition, and resistance to ambient light interference, so it is suitable for acquiring workpiece data in welding scenarios. Its characteristic parameters are shown in Table I.

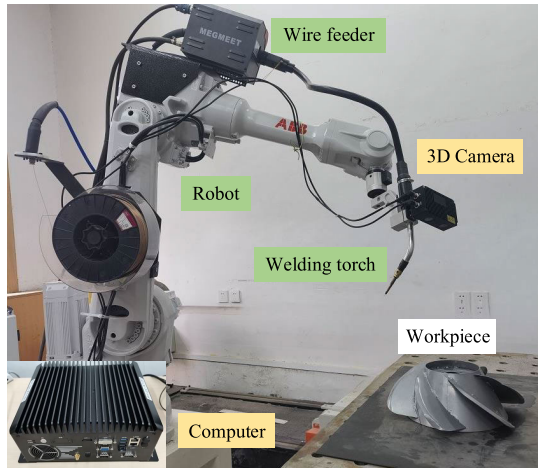


Fig. 10. Welding robot system based on 3D vision.

TABLE I
CHARACTERISTIC PARAMETERS OF THE 3D CAMERA

Case	Parameter	Value
1	Recommended working distance range (mm)	300-600
2	Near-end field of view (mm)	220×160 @ 0.3m
3	Far-end field of view (mm)	430×320 @ 0.6m
4	Resolution	1280×1024
5	Calibration precision (mm)	0.1
6	Typical acquisition time (s)	0.6-1.1
7	Overall dimensions (mm)	145×51×85
8	Weight (kg)	0.7

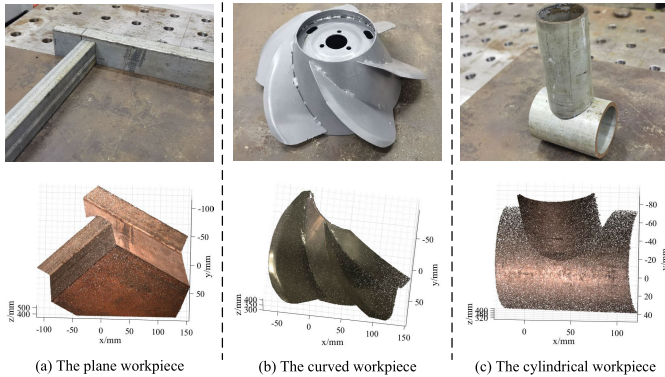


Fig. 11. Welding workpiece models.

The data used in this study were all collected by this 3D camera.

This study is devoted to proposing a feature extraction algorithm suitable for various welding workpieces, so we choose workpieces with different surface curvature changes as the experimental object. The selected workpieces are shown in Fig. 11. Among them, Fig. 11(a) is a part of the side panel of the carriage, and its surface is mainly composed of planes. Fig. 11(b) is an impeller whose surface is mainly composed of curved surfaces. Fig. 11(c) is an orthogonal workpiece model of a circular tube composed of cylindrical surfaces. In this paper, the three kinds of workpieces are called the plane workpiece, the curved workpiece, and the cylindrical workpiece, respectively.

The surface curvatures of the three workpieces are different, and the concave and convexity of the edges are quite different. The generality of the proposed method can be verified by feature extraction of three workpieces. According to the working distance of the camera, by adjusting the robot's pose, the camera can capture the workpiece to obtain a point cloud on the surface of the workpiece. The captured point clouds are shown in Fig. 11, and their units are *mm*.

B. Algorithm Robustness to Parameters

The proposed algorithm primarily involves the configuration of two key parameters: the neighborhood radius for LOBB descriptors and the convergence threshold for K-means clustering.

The neighborhood radius for LOBB descriptors is determined based on the measurement of the workpiece's dimensions. Once established, it seldom requires adjustments. Moreover, the operational environment of the welding robot remains relatively constant, ensuring the stability of the chosen neighborhood radius. Through many experiments, we ultimately settled on a neighborhood radius of $R = 2.5\text{mm}$.

K-means clustering necessitates a convergence threshold to determine when to terminate the iterative process. Generally, the algorithm stops when the average distance between consecutive iterations' cluster centroids falls below the convergence threshold. Based on observations from Fig. 7, it becomes evident that, after normalizing feature vectors, the magnitudes of changes in adjacent features are significantly smaller than 0.001. Therefore, using the plane workpiece model as an example, we deliberated on the selection of convergence threshold values.

By repeating the experiment and taking the average, we can get Table II. It displays the clustering centers corresponding to different orders of magnitude for threshold values ranging from 0.000001 to 0.001. It is apparent that when the threshold value is less than 0.001, the clustering centers remain essentially stable. The large value space of this parameter proves that the performance of the algorithm does not depend much on this parameter. However, it is important to note that smaller threshold values result in longer computation times. Consequently, considering both stability and efficiency, this paper determined a convergence threshold of 0.0001.

C. Evaluation of Feature Extraction

1) *Edge Extraction:* In the experiment, we used the proposed method to extract edge features for the point clouds of the three workpieces. Fig. 12 is the extraction results of the method proposed in this paper. At the same time, this paper selects the edge extraction algorithm proposed by Gumhold [36] and Yang [30] for comparison. Gumhold et al. similarly divide the edges into crease points and boundary points, and Fig. 13 shows the extraction results using the method proposed by Gumhold et al. The method proposed by Yang does not divide the edge of the point cloud into boundary points and crease points. Fig. 14 shows this method's extraction results on three workpieces.

In Fig. 12 and Fig. 13, the blue represents crease points, and the red represents boundary points. From observation, we can draw the following conclusions: For the extraction of boundary points, both methods can be implemented, but the result of

TABLE II
EVALUATION OF CONVERGENCE THRESHOLD VALUE OF K-MEANS

Convergence threshold	Boundary extraction		Crease extraction		Corner extraction	
	Center1	Center2	Center1	Center2	Center1	Center2
0.001	0.6586	0.0224	0.3237	0.0257	0.3708	0.0580
0.0001	0.6551	0.0224	0.3118	0.0248	0.3622	0.0565
0.00001	0.6551	0.0222	0.3144	0.0248	0.3590	0.0560
0.000001	0.6541	0.0222	0.3110	0.0248	0.3611	0.0560

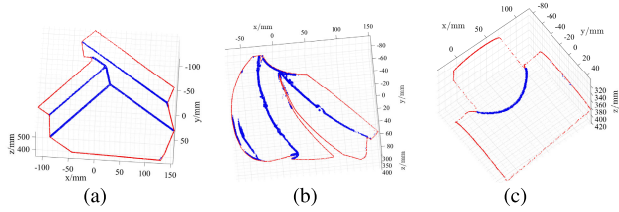


Fig. 12. Edge extraction results of this method.

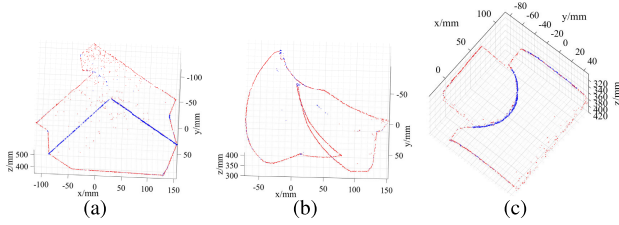


Fig. 13. Edge extraction results of Gumhold's method.

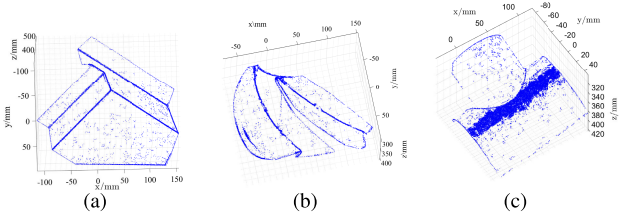


Fig. 14. Edge extraction results of Yang's method.

Gumhold's method is noisier. However, for the extraction of crease points, Gumhold's method loses more than the method in this paper for the plane workpiece and curved workpiece extraction. In addition, in the extraction result of the cylindrical workpiece, the overlapping area of crease points and boundary points extracted by the method in this paper is less, which shows that the method proposed in this paper is more accurate in the feature description.

By comparing Fig. 12 and Fig. 14, we get the following conclusions: Yang's method can extract the crease points and boundary points of the plane workpiece and curved workpiece relatively wholly but cannot extract the crease points of the cylindrical workpiece. Moreover, this method's extraction results are noisier than the method in this paper.

The comprehensive comparison shows that the above three methods can completely extract the workpiece boundary line. However, there is a big difference in the number of crease lines. Therefore, here we quantitatively analyze the extraction of crease lines. First, according to the requirements in the welding scene, we designate the crease line in the point cloud data that is important in describing the geometrical features of the workpiece as the desired crease line. As shown in Fig. 15,

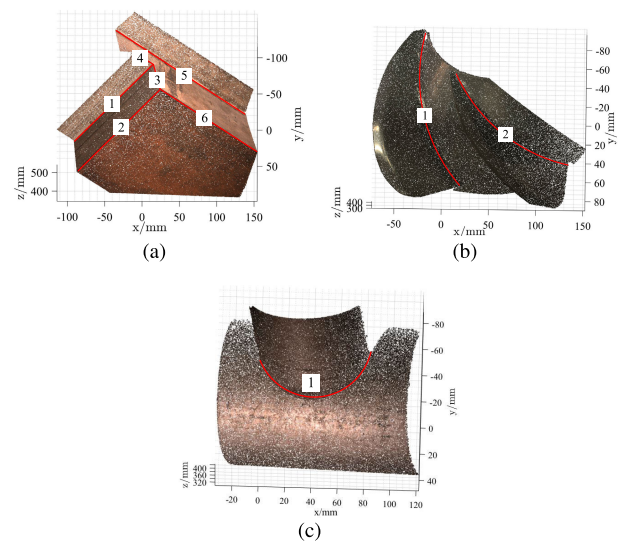


Fig. 15. Desired crease lines marking diagram.

TABLE III
THE NUMBER OF DESIRED CREASE LINES EXTRACTED

Algorithm	The plane workpiece	The curved workpiece	The cylindrical workpiece
The method of this paper	6	2	1
Gumhold's method	2	0	1
Yang's method	6	2	0
The number of marked	6	2	1

we use the drawing tool to mark the desired crease line on the point cloud with red. It can be seen that the number of desired crease lines for the plane workpiece, the curved workpiece, and the cylindrical workpiece are 6, 2, and 1, respectively. Through multiple experiments, we calculate the average number of desired crease lines extracted by the three methods and compare them with the number of markers. The results are shown in the Table III.

As can be seen from Table III, the number of crease lines to be extracted by the method in this paper for the plane workpiece, the curved workpiece, and the cylindrical workpiece is 6, 2, and 1. This is consistent with the number of markers in Fig. 15. The other two methods cannot achieve total extraction. We can conclude that the method proposed in this paper can accurately and low-noise perform the edge extraction of workpieces with different surface curvature changes.

In order to verify the extraction accuracy of the crease lines, we transformed the 9 crease lines into the robot base coordinate system through hand-eye calibration and fitted them. Based on the trajectory curve taught by the robot, the average Euclidean distance error of each crease line is calculated.

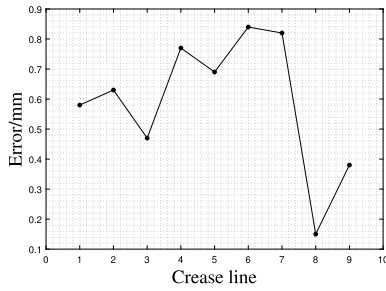


Fig. 16. Error analysis diagram of crease points.

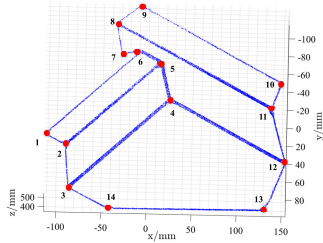


Fig. 17. Desired corners marking diagram.

As shown in Fig. 16, the extraction accuracy of crease points is less than 1 mm, which meets the requirements of welding.

2) *Corner Extraction*: Since the corner features of the plane workpiece are more important to its overall geometric composition, this paper selects the plane workpiece as the experimental verification object for corner extraction. The corner is the part where the edges intersect, so in order to improve the efficiency of the method, the corner extraction is carried out based on the edge line of the workpiece. Fig. 17 is the plane workpiece's edge point cloud extracted by this paper's method. We manually mark the 14 intersecting corner areas in Fig. 17 manually using the drawing software. The marked corner areas are called desired corners and serve as a reference for comparison with the extraction results of the algorithm.

To highlight the advantages of the proposed method, we selected Gumhold's method [36] and the Harris3D algorithm [37] for comparison. Taking the edge point cloud extracted by the method described in this paper as the input point cloud, we conducted multiple experiments and obtained the average results for corner extraction. The final results are shown in Fig. 18, where the corners extracted using the proposed method are represented in red. Specifically, Fig. 18(a) displays the results of the proposed method, Fig. 18(b) shows the results of Gumhold's method, and Fig. 18(c) presents the results of the Harris3D algorithm. The numbers of corner points extracted by the three methods are 11, 3, and 5, respectively.

In Table IV, we call the ratio of the number of corners extracted by different methods to the number of desired corners in Fig. 17 the corner recognition rate, which represents the accuracy of a method for corner feature extraction. At the same time, the average number of points in each corner area is also counted in Table IV. Under the requirement of welding accuracy, the greater the density of point clouds describing the corners, the higher the robustness of the weld extraction method designed on this basis. The corner recognition rate of the method proposed in this paper is 78.57%, and the average

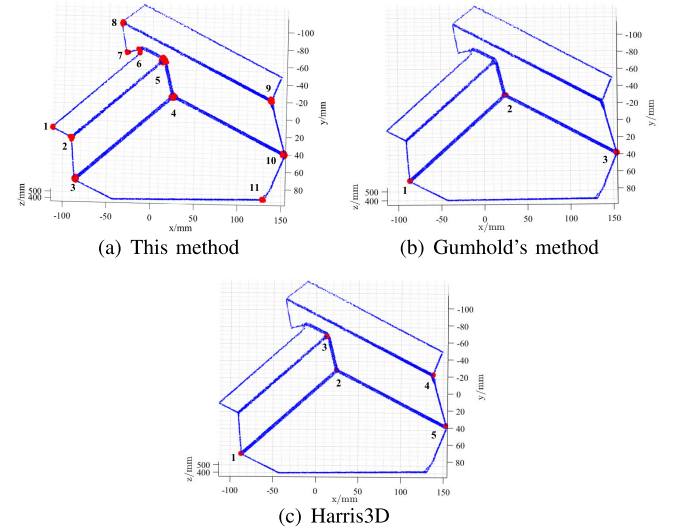


Fig. 18. Comparison of corner extraction results.

TABLE IV
EVALUATION OF CORNER EXTRACTION RESULTS

Algorithm	Corner recognition rate	The average number of points in the corner area
The method of this paper	78.57%	49.46
Gumhold's method	21.43%	23.33
Harris3D	35.71%	1.00

number of points in the corner area is 49.46. Compared with the other two methods, this method is more suitable for corner feature extraction in welding scenes.

D. Advantage Analysis

To verify the advantage of the proposed method in weld extraction application, we carefully selected three methods [3], [16], [21] based on workpiece geometric information corresponding to the corresponding workpiece to perform weld extraction on the three models selected in the experiment. The experimental results are shown in Fig. 19. The red color represents the extracted weld seam. Comparing Fig. 19 with Fig. 12, it is obvious that the welds extracted by these three methods are always included in the crease point extraction results of the proposed method. In addition, from the corner point extraction results in Fig. 18(a), we can observe that effective segmentation of the weld seam can be achieved by positioning corner points. The proposed method can extract three types of workpieces at one time, which can only be achieved by three dedicated methods. Therefore, this proves that the method proposed in this paper has advantages in the application of extracting various welds.

E. Efficiency Analysis

The runtime of the method is an important evaluation metric. In order to facilitate the rapid deployment of the method proposed in the paper, we have chosen the Point Cloud Library (PCL) [38] as the development foundation and have leveraged multi-threading during the programming process to fully utilize hardware resources for improved algorithm efficiency. Our computing platform consists of a

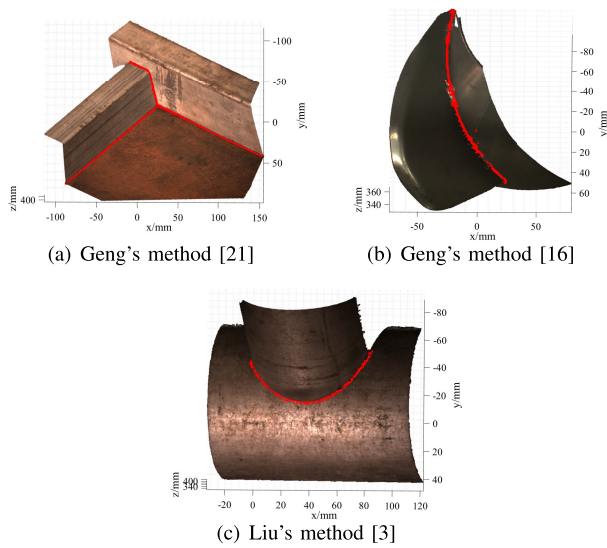


Fig. 19. Weld extraction results of comparative methods.

TABLE V
ALGORITHM RUNNING TIME

Process	Point cloud model	Running time (ms)
Edge extraction	The plane workpiece	3622
	The curved workpiece	3565
	The cylindrical workpiece	2804
Corner extraction	The edge of the plane workpiece	522

computer equipped with an 8-core 2.9GHz CPU and 16GB of RAM.

Through multiple experiments, we calculated the average running time of the proposed algorithm. The specific results are shown in Table V. It can be seen that the overall running time of the algorithm does not exceed 5 s. Compared with the traditional manual teaching and offline programming modes of welding robots, this method has advantages in efficiency.

V. CONCLUSION AND THE FUTURE WORK

For 3D visual welding, this paper proposes an edge and corner point extraction method based on the LOBB. First, the LOBB point cloud feature descriptor is constructed based on OBB. The flatness of LOBB is used to describe the concavity and convexity of the workpiece point cloud, the center offset is used to describe the boundary of the workpiece point cloud, and the corner entropy is used to identify the intersection of edges. Then, to highlight the polarization of the feature distribution, the feature vectors are activated using the $\tanh$ function. Finally, a hierarchical clustering method based on K-means is proposed to realize the extraction of crease points, boundary points, and corner points.

The proposed method is compared with existing methods. The results of edge extraction show that this method can extract the edges of workpieces with various surface curvatures with low noise. The results of corner point extraction show that the corner point recognition of this method has high precision and high density. Its advantages are further verified by comparison with existing weld extraction methods. Therefore, the work of this paper lays the foundation for the universal extraction of a variety of welds.

The current proposed feature extraction algorithm is limited to robotic welding. In future research, we plan to extend and validate this method in other scenarios to more fully explore its applicability and performance.

REFERENCES

- [1] X. Wang, X. Zhou, Z. Xia, and X. Gu, "A survey of welding robot intelligent path optimization," *J. Manuf. Processes*, vol. 63, pp. 14–23, Mar. 2021.
- [2] C. Zheng et al., "Hybrid offline programming method for robotic welding systems," *Robot. Comput.-Integr. Manuf.*, vol. 73, Feb. 2022, Art. no. 102238.
- [3] Y. Liu, J. Liu, and X. Tian, "An approach to the path planning of intersecting pipes weld seam with the welding robot based on non-ideal models," *Robot. Comput.-Integr. Manuf.*, vol. 55, pp. 96–108, Feb. 2019.
- [4] G. Du, M. Chen, C. Liu, B. Zhang, and P. Zhang, "Online robot teaching with natural human-robot interaction," *IEEE Trans. Ind. Electron.*, vol. 65, no. 12, pp. 9571–9581, Dec. 2018.
- [5] J. Cai and T. Lei, "An autonomous positioning method of tube-to-tubesheet welding robot based on coordinate transformation and template matching," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 787–794, Apr. 2021.
- [6] W. Li et al., "Multiple weld seam laser vision recognition method based on the IPCE algorithm," *Opt. Laser Technol.*, vol. 155, Nov. 2022, Art. no. 108388.
- [7] A. Rout, B. B. V. L. Deepak, B. B. Biswal, and G. B. Mahanta, "Weld seam detection, finding, and setting of process parameters for varying weld gap by the utilization of laser and vision sensor in robotic arc welding," *IEEE Trans. Ind. Electron.*, vol. 69, no. 1, pp. 622–632, Jan. 2022.
- [8] W. J. Shao, Y. Huang, and Y. Zhang, "A novel weld seam detection method for space weld seam of narrow butt joint in laser welding," *Opt. Laser Technol.*, vol. 99, pp. 39–51, Feb. 2018.
- [9] J. Fan et al., "An initial point alignment and seam-tracking system for narrow weld," *IEEE Trans. Ind. Informat.*, vol. 16, no. 2, pp. 877–886, Feb. 2020.
- [10] Y. Zhang, T. Liu, B. Li, and Z. Zhang, "Simultaneous monitoring of penetration status and joint tracking during laser keyhole welding," *IEEE/ASME Trans. Mechatronics*, vol. 24, no. 4, pp. 1732–1742, Aug. 2019.
- [11] Y. Gao, P. Zhong, X. Tang, H. Hu, and P. Xu, "Feature extraction of laser welding pool image and application in welding quality identification," *IEEE Access*, vol. 9, pp. 120193–120202, 2021.
- [12] X. Li, X. Li, S. S. Ge, M. O. Khyam, and C. Luo, "Automatic welding seam tracking and identification," *IEEE Trans. Ind. Electron.*, vol. 64, no. 9, pp. 7261–7271, Sep. 2017.
- [13] Z. Wang, "Review of real-time three-dimensional shape measurement techniques," *Measurement*, vol. 156, May 2020, Art. no. 107624.
- [14] R. K. Malhan and S. K. Gupta, "Planning algorithms for acquiring high fidelity pointclouds using a robot for accurate and fast 3D reconstruction," *Robot. Comput.-Integr. Manuf.*, vol. 78, Dec. 2022, Art. no. 102372.
- [15] Z. Wang et al., "Multi-view point clouds registration method based on overlap-area features and local distance constraints for the optical measurement of blade profiles," *IEEE/ASME Trans. Mechatronics*, vol. 27, no. 5, pp. 2729–2739, Oct. 2022.
- [16] Y. Geng, Y. Zhang, X. Tian, X. Shi, X. Wang, and Y. Cui, "A novel welding path planning method based on point cloud for robotic welding of impeller blades," *Int. J. Adv. Manuf. Technol.*, vol. 119, nos. 11–12, pp. 8025–8038, Apr. 2022.
- [17] Z. Lu, J. Fan, Z. Hou, S. Deng, C. Zhou, and F. Jing, "Automatic 3D seam extraction method for welding robot based on monocular structured light," *IEEE Sensors J.*, vol. 21, no. 14, pp. 16359–16370, Jul. 2021.
- [18] P. Zhou, R. Peng, M. Xu, V. Wu, and D. Navarro-Alarcon, "Path planning with automatic seam extraction over point cloud models for robotic arc welding," *IEEE Robot. Autom. Lett.*, vol. 6, no. 3, pp. 5002–5009, Jul. 2021.
- [19] L. Yang and Y. Liu, "A novel 3D seam extraction method based on multi-functional sensor for V-type weld seam," *IEEE Access*, vol. 7, pp. 182415–182424, 2019.
- [20] V. Patil, I. Patil, V. Kalachelvi, and R. Karthikeyan, "Extraction of weld seam in 3D point clouds for real time welding using 5 DOF robotic arm," in *Proc. 5th Int. Conf. Control, Autom. Robot. (ICCAR)*, Apr. 2019, pp. 727–733.

- [21] Y. Geng, M. Lai, X. Tian, X. Xu, Y. Jiang, and Y. Zhang, "A novel seam extraction and path planning method for robotic welding of medium-thickness plate structural parts based on 3D vision," *Robot. Comput.-Integr. Manuf.*, vol. 79, Feb. 2023, Art. no. 102433.
- [22] Y.-J. Cha, K. You, and W. Choi, "Vision-based detection of loosened bolts using the Hough transform and support vector machines," *Autom. Construct.*, vol. 71, pp. 181–188, Nov. 2016.
- [23] Y.-J. Cha, W. Choi, and O. Büyüköztürk, "Deep learning-based crack damage detection using convolutional neural networks," *Comput.-Aided Civil Infrastruct. Eng.*, vol. 32, no. 5, pp. 361–378, Feb. 2017.
- [24] L. Zhu, W. Chen, X. Lin, L. He, and Y. Guan, "Curvature-variation-inspired sampling for point cloud classification and segmentation," *IEEE Signal Process. Lett.*, vol. 29, pp. 1868–1872, 2022.
- [25] S. Wang et al., "A new point cloud simplification method with feature and integrity preservation by partition strategy," *Measurement*, vol. 197, Jun. 2022, Art. no. 111173.
- [26] H. Chen et al., "Multiscale feature line extraction from raw point clouds based on local surface variation and anisotropic contraction," *IEEE Trans. Autom. Sci. Eng.*, vol. 19, no. 2, pp. 1003–1016, Apr. 2022.
- [27] J. Zhou, W. Jin, M. Wang, X. Liu, Z. Li, and Z. Liu, "Improvement of normal estimation for point clouds via simplifying surface fitting," *Comput.-Aided Des.*, vol. 161, Aug. 2023, Art. no. 103533.
- [28] Y. Zhang, X. Wang, Q. Li, and F. Yan, "A new section line extraction method of ring forgings based on normal vector and L1-median," *Measurement*, vol. 184, Nov. 2021, Art. no. 109894.
- [29] D. Li, G. Shi, Y. Wu, Y. Yang, and M. Zhao, "Multi-scale neighborhood feature extraction and aggregation for point cloud segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 6, pp. 2175–2191, Jun. 2021.
- [30] Y. Yang, H. Fang, Y. Fang, and S. Shi, "Three-dimensional point cloud data subtle feature extraction algorithm for laser scanning measurement of large-scale irregular surface in reverse engineering," *Measurement*, vol. 151, Feb. 2020, Art. no. 107220.
- [31] P. Tian, X. Hua, W. Tao, and M. Zhang, "Robust extraction of 3D line segment features from unorganized building point clouds," *Remote Sens.*, vol. 14, no. 14, p. 3279, Jul. 2022.
- [32] S. M. Ahmed, Y. Z. Tan, C. M. Chew, A. A. Mamun, and F. S. Wong, "Edge and corner detection for unorganized 3D point clouds with application to robotic welding," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2018, pp. 7350–7355.
- [33] X. He, R. Wang, C. Feng, and X. Zhou, "A novel type of boundary extraction method and its statistical improvement for unorganized point clouds based on concurrent Delaunay triangular meshes," *Sensors*, vol. 23, no. 4, p. 1915, Feb. 2023.
- [34] W. Ali, S. Abdelkarim, M. Zidan, M. Zahran, and A. El Sallab, "YOLO3D: End-to-end real-time 3D oriented object bounding box detection from LiDAR point cloud," in *Proc. Eur. Conf. Comput. Vis. (ECCV) workshops*, 2018, p. 0.
- [35] N. Chehata, N. David, and F. Bretar, "LiDAR data classification using hierarchical K-means clustering," in *Proc. ISPRS Congr.*, Beijing, China, 2008, vol. 37, no. B3b, pp. 325–330.
- [36] S. Gumhold, X. Wang, and R. MacLeod, "Feature extraction from point clouds," *Proc. 10th Int. Meshing Roundtable*, 2001, pp. 293–305.
- [37] I. Sipiran and B. Bustos, "Harris 3D: A robust extension of the Harris operator for interest point detection on 3D meshes," *Vis. Comput.*, vol. 27, no. 11, pp. 963–976, Nov. 2011.
- [38] R. B. Rusu and S. Cousins, "3D is here: Point cloud library (PCL)," in *Proc. IEEE Int. Conf. Robot. Autom.*, May 2011, pp. 1–4.



Yusen Geng was born in Qingdao, China, in 1993. He received the M.Eng. degree in control science and engineering from Jinan University, Jinan, China, in 2020. He is currently pursuing the Ph.D. degree in pattern recognition and intelligent systems with the School of Control Science and Engineering, Shandong University, Jinan. His current research interests include industrial robotics and machine vision.



Xincheng Tian was born in Dezhou, China, in 1965. He received the B.Eng. and M.Eng. degrees in industrial automation from Shandong Industrial University (now Shandong University), Jinan, China, in 1988 and 1993, respectively, and the Ph.D. degree in guidance, control, and simulation from the Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2000.

He is currently a Professor with the School of Control Science and Engineering, Shandong University. His main researches include robot control technology, the intelligent control of the electromechanical systems, and computer control technology and application.



Fuquan Zheng was born in Dezhou, China, in 1966. He received the B.Eng. and M.Eng. degrees in mechanical manufacturing and design from Shandong Industrial University (now Shandong University), Jinan, China, in 1988 and 1991, respectively. He is currently a Professor with the Institute of Automation, Shandong Academy of Sciences, Qilu University of Technology (Shandong Academy of Sciences). His main research interests include instrumentation and detection, industrial intelligence, and information security.



Yong Jiang was born in Lu'An, China, in 1999. He received the B.Eng. degree from Anhui University, Hefei, China, in 2021. He is currently pursuing the M.Eng. degree in control engineering with the School of Control Science and Engineering, Shandong University, Jinan, China. His current research interests include industrial robotics and machine vision.



Yuankai Zhang was born in Puyang, China, in 1996. He received the M.Eng. degree in control science and engineering from Shandong University, Jinan, China, in 2022, where he is currently pursuing the Ph.D. degree in pattern recognition and intelligent systems. His current research interests include industrial robotics and machine vision.



Min Lai was born in Ganzhou, China, in 1998. He received the B.Eng. degree from Shandong University, Jinan, China, in 2021, where he is currently pursuing the M.Eng. degree in control science and engineering. His current research interests include industrial robotics and machine vision.